Gadget Mapping GUI – Email Notifications (v14)

Functional & Technical Documentation

Overview

The Gadget Mapping GUI includes an optional email notification system designed to provide clear operational visibility for automated inbox processing. When enabled, the system sends structured email updates for successful processing and failure events. Email functionality is non-blocking and never interrupts core workflows.

Configuration

Email behaviour is configured via the inbox configuration file located at:

`C:\GadgetClient\in\inbox_config.json`

Key configuration fields include SMTP connection details, sender and recipient addresses, enabled events, and subject prefixes. Configuration is validated at runtime. Invalid or missing settings disable email sending without stopping processing.

Supported Email Events

- Success – Triggered when a file completes processing successfully.
- Failure – Triggered when processing fails and a failure pack is generated.

Success Email Behaviour

A success email is sent only after Workflow 1 completes, any enabled Workflow 2 outputs are written, bordereau outputs are generated (if enabled), and the source file is moved to the Processed folder.

- Resolved client name
- Original filename
- Invoice month
- Script version
- Base output directory
- Clean CSV output
- Workflow 2 outputs (if enabled)
- Per-client bordereau files (if enabled)
- Master bordereau file (if enabled)

Failure Email Behaviour

Failure emails are triggered when processing halts due to validation errors, missing mappings, encoding issues, strict value rule violations, or unexpected runtime exceptions.

- Best-effort client name

- Original filename
- Invoice month
- Failure category
- Failure reason
- Path to the generated failure pack

Client Name Resolution (v14)

Email notifications use the best available client name, resolved in the following order:

- Client name inferred from the processed CSV
- Client name resolved via client_db_v15 (if available)
- Fallback to inferred or UNKNOWN_CLIENT

Reliability & Safety Guarantees

- Email functionality is fully optional.
- SMTP or email failures never interrupt processing.
- All email operations are wrapped in defensive try/except blocks.
- Locked builds remain stable even if email or client database components are missing.

Email Notification Flow Diagram

```
[Inbox File]
|
v
[Workflow 1 – Mapping]
|
v
[Validation]
|
+---- FAIL ----> [Failure Pack Created] --> [Failure Email Sent]
|
v
[Workflow 2 (Optional)]
|
v
[Bordereau (Optional)]
|
v
[Move to Processed]
|
v
[Success Email Sent]
```

Summary

The email notification system in Gadget Mapping GUI v14 provides reliable, human-readable operational feedback for automated processing. It enhances visibility while maintaining strict separation from core processing logic, making it suitable for unattended and production-style workflows.